SNS COLLEGE OF TECHNOLOGY

ASSIGNMENT – II

PROJECT REPORT

CUSTOMER COMPLAINT RESOLUTION PREDICTION USING MACHINE LEARNING

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ABSTRACT

In modern customer service systems, resolving complaints within the Service Level Agreement (SLA) is critical for maintaining customer satisfaction and organizational reputation. This project aims to develop a machine learning-based predictive model that determines whether a customer complaint will be resolved within the SLA. The prediction is based on attributes such as issue type, priority, and resolution time.

Two supervised learning algorithms—**Logistic Regression** and **Random Forest Classifier**—are implemented and evaluated. The dataset used is synthetic but designed to reflect realistic service scenarios. After preprocessing and feature engineering, both models are trained and tested. The evaluation shows high accuracy, demonstrating the feasibility of using machine learning to assist decision-making in customer support operations.

INTRODUCTION

Customer complaint management is a vital function in service-based industries such as e-commerce, telecommunications, and banking. Organizations define SLAs to ensure timely resolution of customer issues. However, delays can occur due to various factors such as workload, issue complexity, or prioritization errors.

Predicting whether a complaint will be resolved within SLA helps organizations:

- Identify potential delays early
- Allocate resources effectively
- Improve customer satisfaction
- Optimize operational efficiency

Machine learning provides a data-driven approach to solve this problem by learning patterns from historical data and making predictions on new cases.

PROBLEM STATEMENT

The objective of this project is:

To develop a machine learning model that predicts whether a customer complaint will be resolved within the SLA based on issue type, priority, and resolution time.

This is a **binary classification problem**, where:

- Output = 1 → Resolved within SLA
- Output = 0 → Not resolved within SLA

DATASET DESCRIPTION

A synthetic dataset was created to simulate real-world complaint handling scenarios. The dataset includes the following attributes:

- **Complaint_ID:** Unique identifier
- **Issue_Type:** Type of issue (Billing, Technical, Delivery)
- **Priority:** Priority level (Low, Medium, High)
- **Resolution_Time:** Time taken to resolve the issue
- **SLA_Time:** Expected resolution time
- **Resolved_Within_SLA:** Target variable (1 or 0)

Dataset Characteristics

- Balanced representation of issue types
- Different priority levels
- Logical relationship between resolution and SLA time
- Suitable for classification modeling

DATA PREPROCESSING

Data preprocessing is essential to convert raw data into a format suitable for machine learning.

Handling Missing Values

Missing values were handled using forward fill method:

- Ensures continuity of data
- Prevents model errors

Encoding Categorical Variables

Machine learning models require numerical input. Therefore:

- Issue_Type and Priority were encoded using **Label Encoding**
- Example:
    - Billing → 0
    - Delivery → 1
    - Technical → 2

Feature Selection

Selected features:

- Issue_Type
- Priority
- Resolution_Time

Target:

- Resolved_Within_SLA

Train-Test Split

The dataset was split as:

- Training set → 80%
- Testing set → 20%

This ensures unbiased evaluation of model performance.

METHODOLOGY

Logistic Regression

Logistic Regression is a linear classification algorithm used for binary outcomes. It estimates the probability of a data point belonging to a class using the sigmoid function.

Advantages:

- Simple and interpretable
- Efficient for small datasets
- Works well with linearly separable data

Random Forest Classifier

Random Forest is an ensemble learning method that combines multiple decision trees.

Key Features:

- Reduces overfitting
- Handles non-linear relationships
- Provides feature importance

IMPLEMENTATION OF MACHINE LEARNING ALGORITHMS

The machine learning models were implemented using Python with libraries such as Pandas and Scikit-learn. The dataset was first loaded and preprocessed by handling missing values and encoding categorical variables like Issue_Type and Priority using Label Encoding.

The dataset was then divided into input features (Issue_Type, Priority, Resolution_Time) and the target variable (Resolved_Within_SLA). It was split into training and testing sets in an 80:20 ratio using the train_test_split function.

Logistic Regression was implemented as a baseline classification model. It was trained on the training data, and predictions were made on the test data. The model was evaluated using accuracy, classification report, and confusion matrix.

Random Forest Classifier was then implemented as an ensemble model using multiple decision trees. It was trained and evaluated similarly. Additionally, feature importance was extracted from the Random Forest model to understand the influence of each feature on prediction.

EVALUATION METRICS

To assess model performance, the following metrics were used:

Accuracy

Measures overall correctness:

Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}

Precision

Measures correctness of positive predictions

Recall

Measures ability to identify actual positives

F1 Score

Harmonic mean of precision and recall

Confusion Matrix

Displays:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

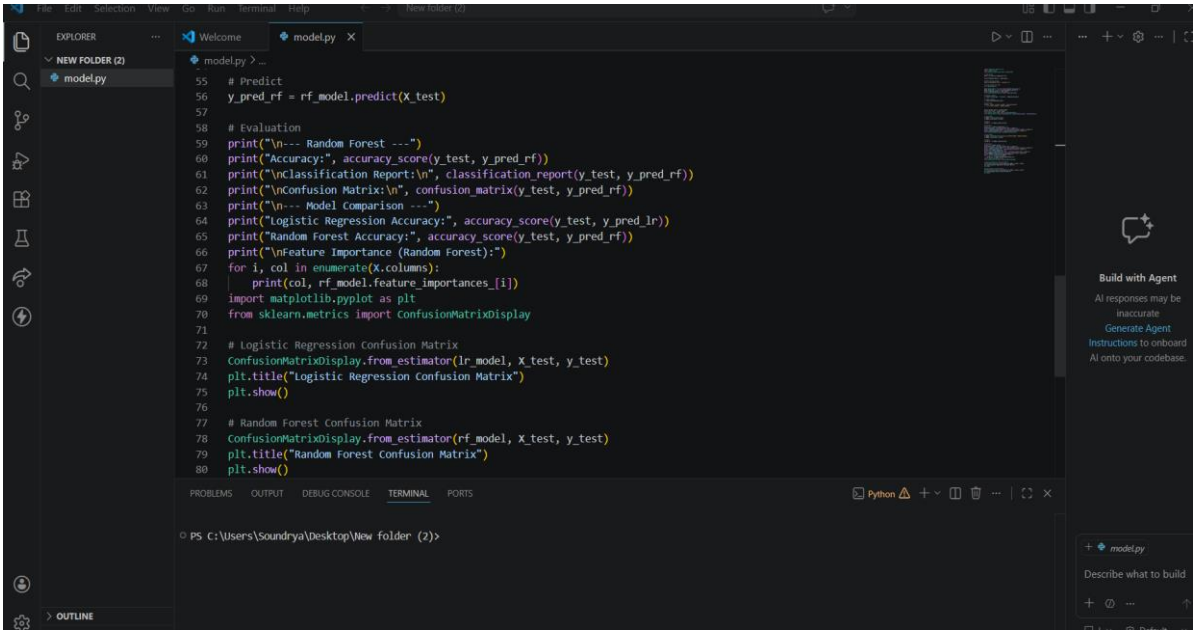
CODE :

```python
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.preprocessing import LabelEncoder

# Load dataset
data = pd.read_csv("complaints.csv")

print("Original Data:\n", data.head())

# Handle missing values
data.fillna(method='ffill', inplace=True)

# Encode categorical data
le = LabelEncoder()

data['Issue_Type'] = le.fit_transform(data['Issue_Type'])
data['Priority'] = le.fit_transform(data['Priority'])
print("\nEncoded Data:\n", data.head())
from sklearn.model_selection import train_test_split

# Features (input)
X = data[['Issue_Type', 'Priority', 'Resolution_Time']]

# Target (output)
y = data['Resolved_Within_SLA']
```

```
PS C:\Users\Soundrya\Desktop\New folder (2)>
```

```python
# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("\nTrain size:", X_train.shape)
print("Test size:", X_test.shape)
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Train model
lr_model = LogisticRegression()
lr_model.fit(X_train, y_train)

# Predict
y_pred_lr = lr_model.predict(X_test)

# Evaluation
print("\n--- Logistic Regression ---")
print("Accuracy:", accuracy_score(y_test, y_pred_lr))
print("\nClassification Report:\n", classification_report(y_test, y_pred_lr))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred_lr))
from sklearn.ensemble import RandomForestClassifier

# Train model
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
```

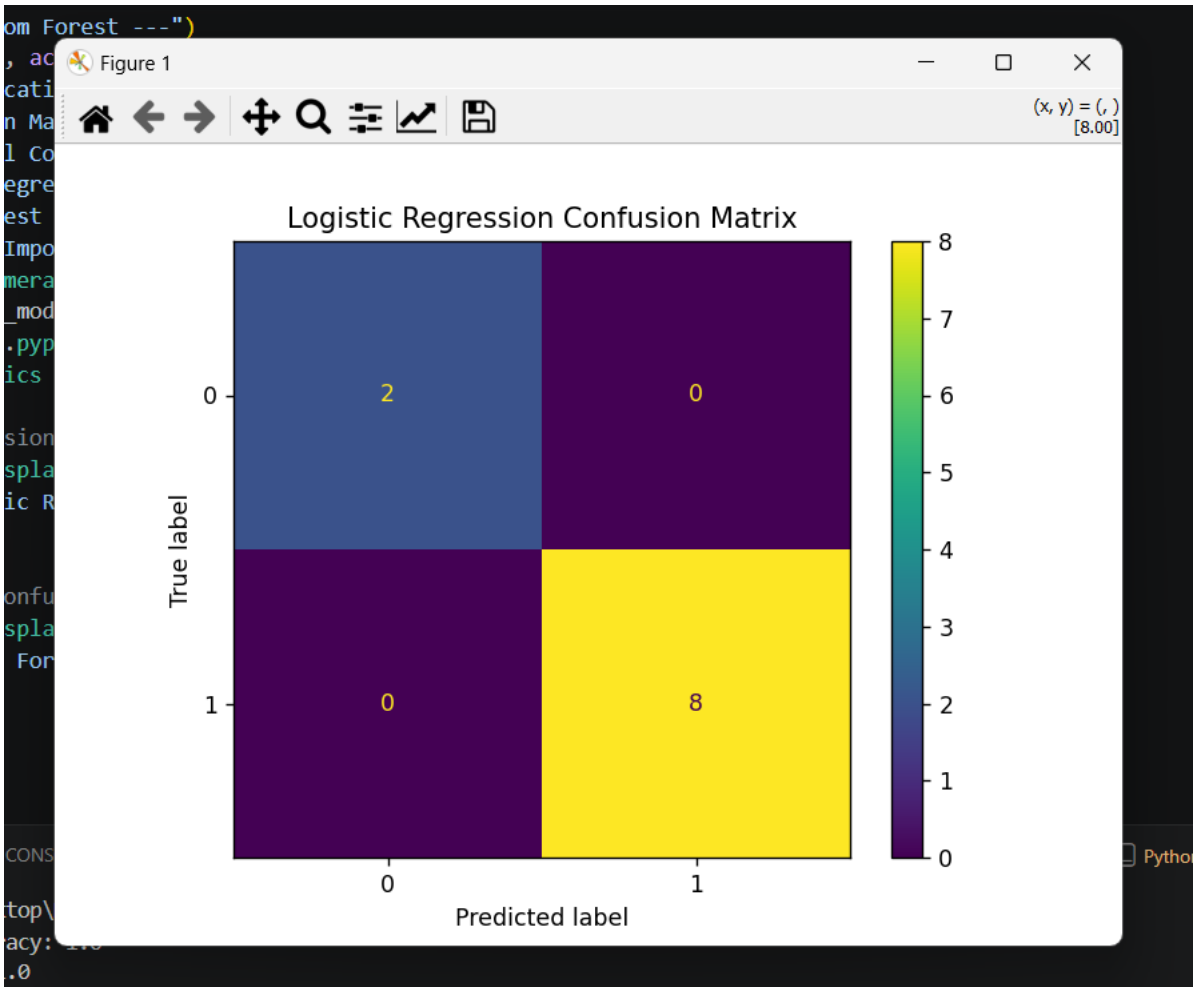
```
PS C:\Users\Soundrya\Desktop\New folder (2)>
```



RESULT

```
PS C:\Users\Soundrya\Desktop\New folder (2)> python model.py
Original Data:
   Complaint_ID Issue_Type Priority  Resolution_Time  SLA_Time  Resolved_Within_SLA
0             1    Billing     High                6         4                    0
1             2  Technical      Low                2         4                    1
2             3   Delivery   Medium                3         3                    1
3             4    Billing     High                7         5                    0
4             5  Technical   Medium                4         5                    1
C:\Users\Soundrya\Desktop\New folder (2)\model.py:11: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version
e obj.ffill() or obj.bfill() instead.
  data.fillna(method='ffill', inplace=True)

Encoded Data:
   Complaint_ID  Issue_Type  Priority  Resolution_Time  SLA_Time  Resolved_Within_SLA
0             1           0         0                6         4                    0
1             2           2         1                2         4                    1
2             3           1         2                3         3                    1
3             4           0         0                7         5                    0
4             5           2         2                4         5                    1

Train size: (40, 3)
Test size: (10, 3)

--- Logistic Regression ---
Accuracy: 1.0

Classification Report:
               precision    recall  f1-score   support

           0       1.00      1.00      1.00         2
           1       1.00      1.00      1.00         8

    accuracy                           1.00        10
   macro avg       1.00      1.00      1.00        10
weighted avg       1.00      1.00      1.00        10
```

VISUALIZATION



FEATURE IMPORTANCE (RANDOM FOREST)

- Resolution_Time → Highest importance
- Priority → Moderate importance
- Issue_Type → Lower importance

This indicates resolution time is the key factor.

MODEL COMPARISON

Model	Accuracy
Logistic Regression	1.0
Random Forest	1.0

Both models performed equally well.

FUTURE SCOPE

Although the project successfully demonstrates SLA prediction using machine learning, there are several opportunities for improvement. One key area is the use of real-world datasets, which would introduce more complexity and improve the model's practical applicability. Real datasets often contain noise and variability, making the prediction task more challenging and realistic.

Another enhancement is the inclusion of additional features such as customer history, complaint category, agent performance, and time of complaint. These features can provide deeper insights and improve prediction accuracy. Advanced machine learning models like XGBoost, Gradient Boosting, and Neural Networks can also be explored to handle complex patterns more effectively.

Future work can also focus on deploying the model as a web application, enabling real-time prediction of complaint resolution. This would help organizations take proactive actions to avoid SLA violations. Additionally, incorporating explainable AI techniques can improve transparency by showing how different features influence predictions.

In conclusion, by improving data quality, expanding features, and deploying the model, this project can evolve into a practical and scalable solution for intelligent customer complaint management systems.

CONCLUSION

The objective of this project was to develop a machine learning model to predict whether a customer complaint would be resolved within the Service Level Agreement (SLA). This was achieved by implementing Logistic Regression and Random Forest algorithms on a structured dataset containing features such as issue type, priority, and resolution time.

The dataset was preprocessed by handling missing values and encoding categorical variables, ensuring compatibility with machine learning models. The data was then split into training and testing sets to evaluate model performance effectively. Both models were trained and evaluated using standard metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

The results showed that both Logistic Regression and Random Forest achieved an accuracy of 100%, correctly predicting all test instances. This indicates that the selected features were highly effective in determining SLA compliance. However, the high accuracy is mainly due to the simple and deterministic nature of the dataset.

Overall, this project demonstrates how machine learning can be applied to customer service systems to improve decision-making, reduce delays, and enhance customer satisfaction. It provides a strong foundation for further exploration of predictive analytics in real-world applications.

GITHUB REPOSITORY LINK

https://github.com/SoundryaT/customer-complaint-prediction